

Security and Protection of Systems

Lecture 07 *Introduction to Cryptography*

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ITNM3312, Security & Protection of Systems

Outlines

- *Introduction*
- *Cryptographic Transformations*
- *Stream and Block Cipher*
- *Feistel cipher*
- *Data Encryption Standard (DES)*

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Terminologies

- Plaintext: *An original message*
- Ciphertext: *The coded message*
- Deciphering/decryption: *Restoring the plaintext from the ciphertext*
- Cryptography: *The area of study of the many schemes used for encryption*
- Cryptographic system/cipher: *A scheme being use to convert from ciphertext to plaintext*
- Cryptanalysis: *Techniques used for deciphering a message without any knowledge of the enciphering details*
- Cryptology: *The areas of cryptography and cryptanalysis*

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Characteristics

1. The type of operations used for transforming plaintext to ciphertext
 - Substitution
 - Transposition
2. The number of keys used
 - Symmetric, single-key, secret-key, conventional encryption
 - Asymmetric, two-key, or public-key encryption
3. The way in which the plaintext is processed
 - Block cipher
 - Stream cipher

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Encryption Scheme Security

- Unconditionally secure: No matter how much time an opponent has, it is impossible to decrypt the ciphertext simply because the required information is not there
- Computationally secure: The cost of breaking cipher exceeds the value of encrypted information, The time required to break the cipher exceeds the useful lifetime of information
- Diffusion: The statistical structure of plaintext is dissipated into long-range statistics of the ciphertext. This is achieved by having each plaintext digit affect the value of many ciphertext digits
- Confusion: Seeks to make the relationship between the statistics of the ciphertext and the value of the encryption key as complex as possible . Even if the attacker can get some handle on the statistics of the ciphertext, the way in which the key was used to produce that ciphertext is so complex as to make it difficult to deduce the key

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Cryptographic Transformations

1- Substitutions: Each plaintext element or group of elements is uniquely replaced by a corresponding ciphertext element or group of elements

- Examples of Substitution techniques such as *Caesar* - *Playfair* - *Vigenère*.

2- Permutation (Transposition) : No elements are added or deleted or replaced in the sequence, rather the order in which the elements appear in the sequence is changed

- Examples of permutation techniques Such as *Rail Fence Cipher*, *Row Transposition Cipher*

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1- Caesar Cipher

- Simplest and earliest known use of a **substitution** cipher used by **Julius Caesar**
- Replacing** each letter of the alphabet with the letter **standing three places** further down the alphabet
- Mathematically: the general Caesar algorithm is:

$$C = E(k, p) = (p + k) \bmod 26, P = D(k, C) = (C - k) \bmod 26$$

- Where k takes on a value in the range 1 to 25.
- Example: **plain:** meet me after the toga party

cipher: PHHW PH DIWHU WKH WRJD SDUWB

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2- Playfair Cipher

- A **substitution** cipher used **replacing** each couple of the plaintext by diagrams in 5x5 matrix as follows:

- Create 5 x 5 matrix of letters using a keyword such as "MONARCHY"
- Create a diagram by splitting plaintext to couple of letters

M	O	N	A	R
C	H	Y	B	D
E	F	G	I / J	K
L	P	Q	S	T
U	V	W	X	Z

- In the case of the last letter is **single**, couple it by letter **x**.
- If a two letters are **repeated**, insert letter **x** in the **middle**.

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2- Playfair Cipher

- If the letters were in different rows and columns, encrypts them by rectangle circular left/right swap
- If the letters were in the same rows, encrypts them by circular left swap.
- If the letters were in the same column, encrypts them by circular down swap.
- Reverse the last steps to decrypt the message

M	O	N	A	R
C	H	Y	B	D
E	F	G	I / J	K
L	P	Q	S	T
U	V	W	X	Z

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2- Playfair Cipher

- Example 1 : Plaintext: **attack**
- Diagram : at ta ck
- Repeating: no repeating
- at encrypted to RS
- Ta encrypted to SR
- Ck encrypted to DE.
- Ciphertext: RSSRDE
- Example 2 : Plaintext: **balloon**
- Diagram : ba ll oo n
- Repeating: yes: ba lx lo on
- ba same column encrypt to IB
- lx encrypted to SU
- lo encrypted to PM.
- on same row encrypted to NA
- Ciphertext: RSSRDE

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3- Rail Fence Cipher

- Simplest **transposition** cipher
- Plaintext is written down as a sequence of diagonals and then read off as a sequence of rows
- To encipher message "meet me after the toga party" with a rail fence of depth 2, we would write:
 - m e m a t r h t g p r y
 - e t e f e t e o a a t
- Encrypted message is: **MEMATRHTGPRYETEFETEOAAT**

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4- Row Transposition Cipher

- Is a more complex **transposition**, Write message in a rectangle, row by row, and read the message off, column by column, but permute the order of the columns
- The order of the columns then becomes the key to the algorithm
- Key: 3 4 1 2 5 6 7
- Plaintext: a t t a c k p

o s t p o n e

d u n t i l t

w o a m x y z

Ciphertext: TTNAAPTMTSUOAODWCOIXKNLYPETZ

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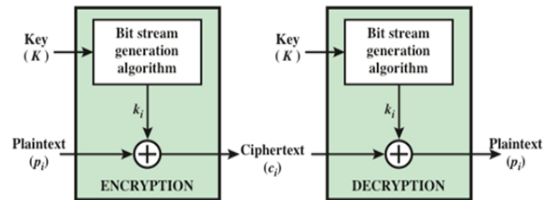
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Stream Cipher

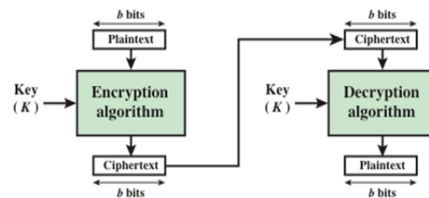
- Encrypts data stream one bit or one byte at a time
- In the ideal case, the **keystream** is as long as the plaintext bit stream
- **Keystream** must be provided to both users via some independent and secure channel; introduces insurmountable logistical problems if intended data traffic is very large
- bit-stream generator must be implemented as an algorithmic procedure so that **the cryptographic bit stream can be produced by both users**
- It must be computationally impractical to predict future portions of the bit stream based on previous portions of the bit stream
- The two users need only share the generating key and each can produce the keystream

Block Cipher

- A block of plaintext is treated as a whole and used to produce a ciphertext block of equal length
- Typically a block size of 64 or 128 bits is used
- As with a stream cipher, the two users share a symmetric encryption key
- The majority of network-based symmetric cryptographic applications make use of block ciphers



(a) Stream Cipher Using Algorithmic Bit Stream Generator



(b) Block Cipher

Block Cipher Design Principles

- 1- Number of Rounds:** The greater number of rounds, the more difficult it is to perform cryptanalysis.
- 2- Design of Function F (Feistel):** The more nonlinear F, the more difficult any type of cryptanalysis will be. **Strict avalanche criterion (SAC):** O/P bits of an S-box should change with probability 1/2 when any single I/P bit is inverted and **Bit independence criterion (BIC):** O/P bits should change independently when any single I/P bit is inverted.
- 3- Key Schedule Algorithm:** the key is used to generate one subkey for each round. In general, we would like to select subkeys to maximize the difficulty of deducing individual subkeys and the difficulty of working back to the main key. It is suggested that, at a minimum, the key schedule should guarantee key/ciphertext Strict Avalanche Criterion and Bit Independence Criterion

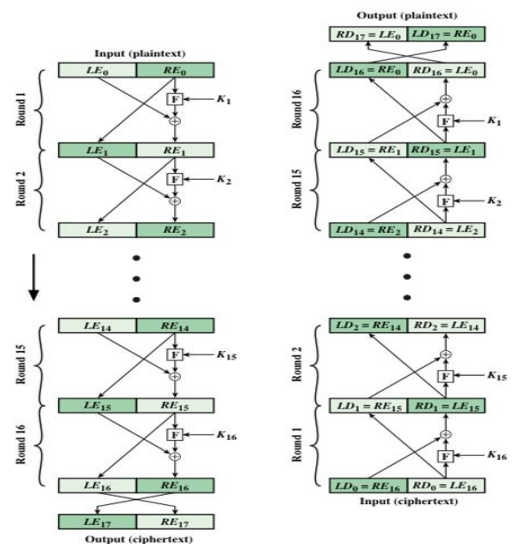
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Feistel Cipher

- Feistel proposed the use of a cipher that alternates **substitutions** and **permutations**
- Is a practical application of a proposal by Claude Shannon to develop a product cipher that alternates confusion and diffusion functions
- Is the structure used by many significant symmetric block ciphers currently in use



Feistel Cipher Design Features

1. **Block size:** Larger block sizes mean greater security but reduced speed
2. **Key size:** Larger key size means greater security but may decrease speeds
3. **Number of rounds:** The essence of the Feistel cipher is that a single round offers inadequate security but that multiple rounds offer increasing security
4. **Subkey generation algorithm:** Greater complexity in this algorithm should lead to greater difficulty of cryptanalysis
5. **Round function F:** Greater complexity generally means greater resistance to cryptanalysis

Feistel Cipher Design Features

6. **Fast software encryption/decryption:** In many cases, encrypting is embedded in applications or utility functions, preclude a hardware implementation; accordingly, the speed of execution of the algorithm becomes a concern
7. **Ease of analysis :** If the algorithm can be concisely and clearly explained, it is easier to analyze that algorithm for cryptanalytic vulnerabilities and therefore develop a higher level of assurance as to its strength

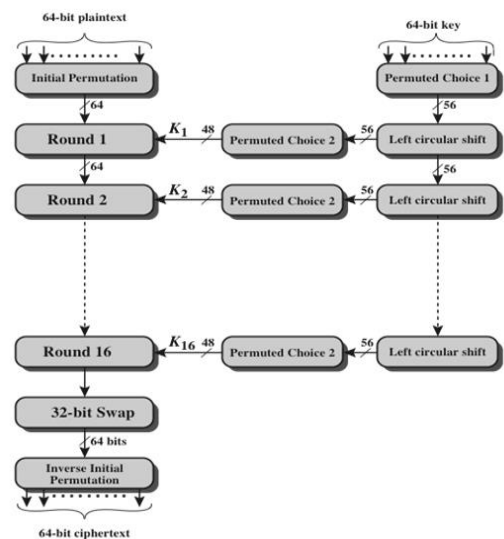
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Data Encryption Standard (DES)

- Issued in 1977 by NIST as Federal Information Processing Standard
- The most widely used encryption scheme until introduction of AES in 2001
- Data are encrypted in 64-bit blocks using a 56-bit key
- The algorithm transforms 64-bit input in a series of steps into a 64-bit output
- The same steps, with the same key, are used to reverse the encryption



DES Example

Round	<i>K_i</i>	<i>L_i</i>	<i>R_i</i>
IP		5a005a00	3cf03c0f
1	1e030f03080d2930	3cf03c0f	bad22845
2	0a31293432242318	bad22845	99e9b723
3	23072318201d0c1d	99e9b723	0bae3b9e
4	05261d3824311a20	0bae3b9e	42415649
5	3325340136002c25	42415649	18b3fa41
6	123a2d0d04262a1c	18b3fa41	9616fe23
7	021f120b1c130611	9616fe23	67117cf2
8	1c10372a2832002b	67117cf2	c11bfc09
9	04292a380c341f03	c11bfc09	887fbc6c
10	2703212607280403	887fbc6c	600f7e8b
11	2826390c31261504	600f7e8b	f596506e
12	12071c241a0a0f08	f596506e	738538b8
13	300935393c0d100b	738538b8	c6a62c4e
14	311e09231321182a	c6a62c4e	56b0bd75
15	283d3e0227072528	56b0bd75	75e8fd8f
16	2921080b13143025	75e8fd8f	25896490
IP-1		da02ce3a	89ecac3b

Avalanche Effect in DES: Change in Plaintext

Round		δ
	02468aceeca86420 12468aceeca86420	1
1	3cf03c0fbad22845 3cf03c0fbad32845	1
2	bad2284599e9b723 bad3284539a9b7a3	5
3	99e9b7230bae3b9e 39a9b7a3171cb8b3	18
4	0bae3b9e42415649 171cb8b3ccaca55e	34
5	4241564918b3fa41 ccaca55ed16c3653	37
6	18b3fa419616fe23 d16c3653cf402c68	33
7	9616fe2367117cf2 cf402c682b2cefbcb	32
8	67117cf2c11bfc09 2b2cefbcb99f91153	33

Round		δ
9	c11bfc09887fbc6c 99f911532eed7d94	32
10	887fbc6c600f7e8b 2eed7d94d0f23094	34
11	600f7e8bf596506e d0f23094455da9c4	37
12	f596506e738538b8 455da9c47f6e3cf3	31
13	738538b8c6a62c4e 7f6e3cf34bc1a8d9	29
14	c6a62c4e56b0bd75 4bc1a8d91e07d409	33
15	56b0bd7575e8fd8f 1e07d4091ce2e6dc	31
16	75e8fd8f25896490 1ce2e6dc365e5f59	32
IP-1	da02ce3a89ecac3b 057cde97d7683f2a	32

Avalanche Effect in DES: Change in Key

Round		δ	Round		δ
	02468aceeca86420 02468aceeca86420	0	9	c11bfc09887fbc6c 548f1de471f64dfd	34
1	3cf03c0fbad22845 3cf03c0f9ad628c5	3	10	887fbc6c600f7e8b 71f64dfd4279876c	36
2	bad2284599e9b723 9ad628c59939136b	11	11	600f7e8bf596506e 4279876c399fdc0d	32
3	99e9b7230bae3b9e 9939136b768067b7	25	12	f596506e738538b8 399fdc0d6d208dbb	28
4	0bae3b9e42415649 768067b75a8807c5	29	13	738538b8c6a62c4e 6d208dbbb9bdeaaa	33
5	4241564918b3fa41 5a8807c5488dbe94	26	14	c6a62c4e56b0bd75 b9bdeeaad2c3a56f	30
6	18b3fa419616fe23 488dbe94aba7fe53	26	15	56b0bd7575e8fd8f d2c3a56f2765c1fb	33
7	9616fe2367117cf2 aba7fe53177d21e4	27	16	75e8fd8f25896490 2765c1fb01263dc4	30
8	67117cf2c11bfc09 177d21e4548f1de4	32	IP-1	da02ce3a89ecac3b ee92b50606b62b0b	30

Thanks